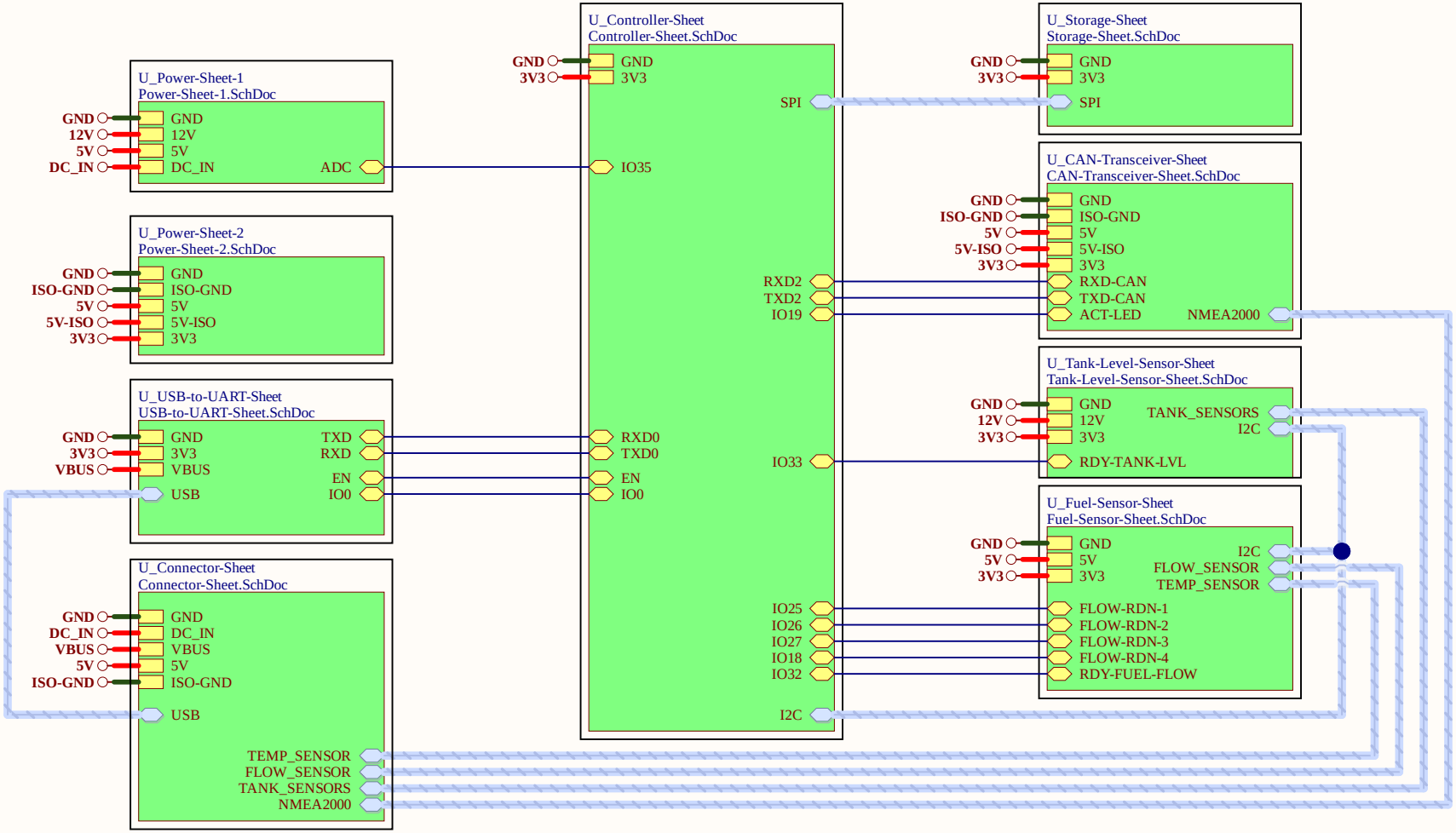
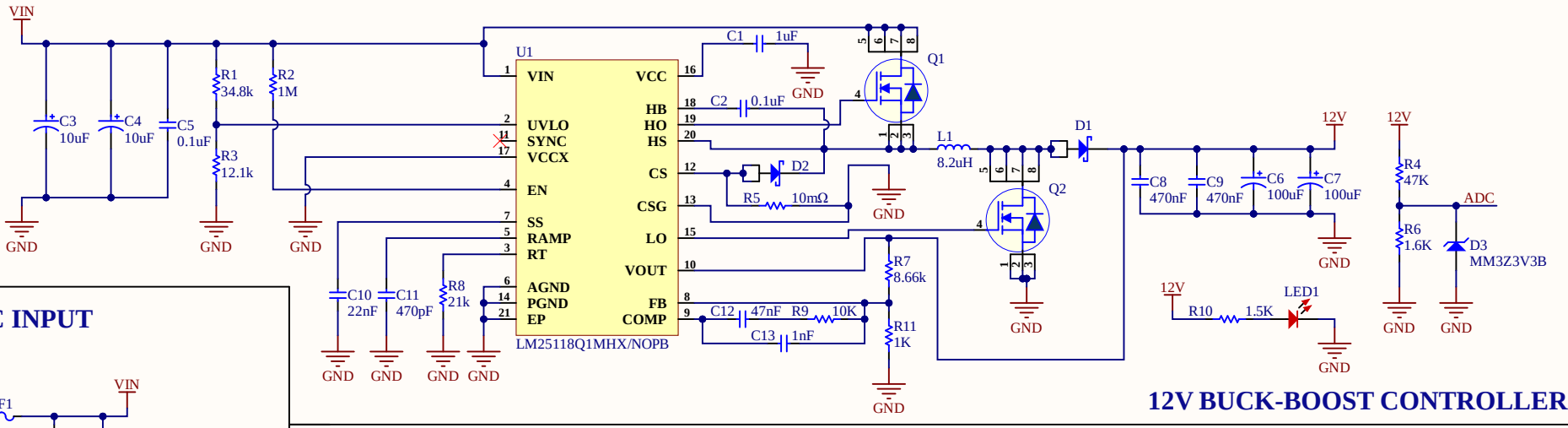


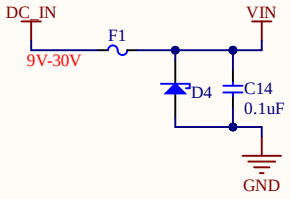
eSpark Consulting Group

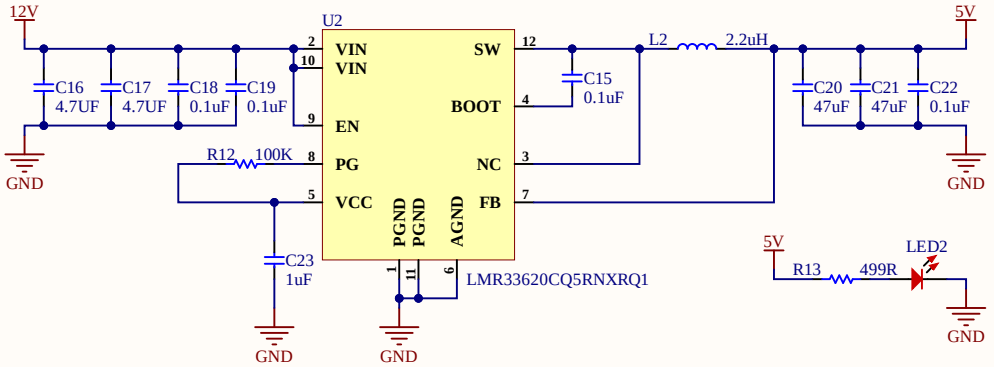
Title: Top-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 10



12V BUCK-BOOST CONTROLLER

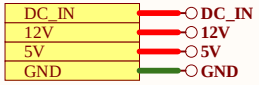
DC INPUT





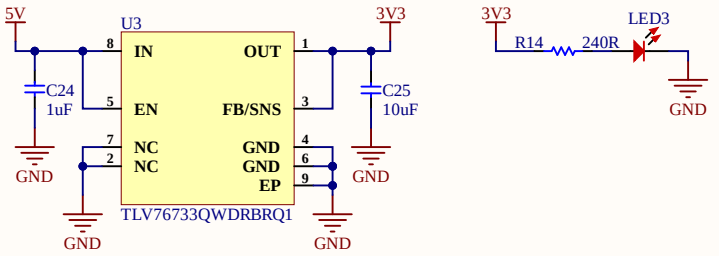
5V BUCK CONVERTER



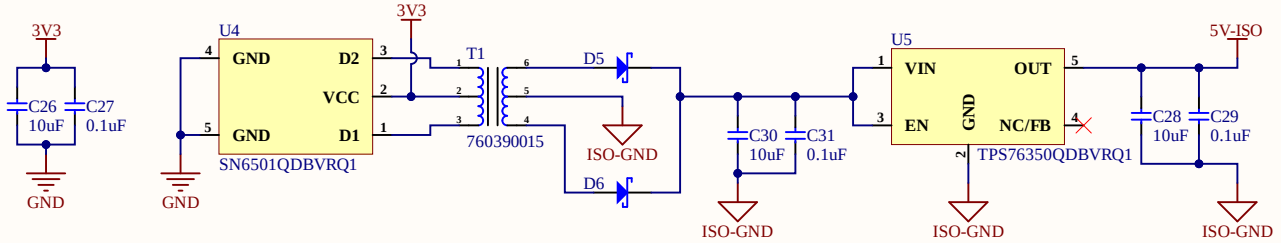


eSpark Consulting Group

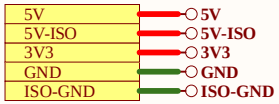
Title: Power-Sheet-1.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 09/05/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 10



3.3V LDO

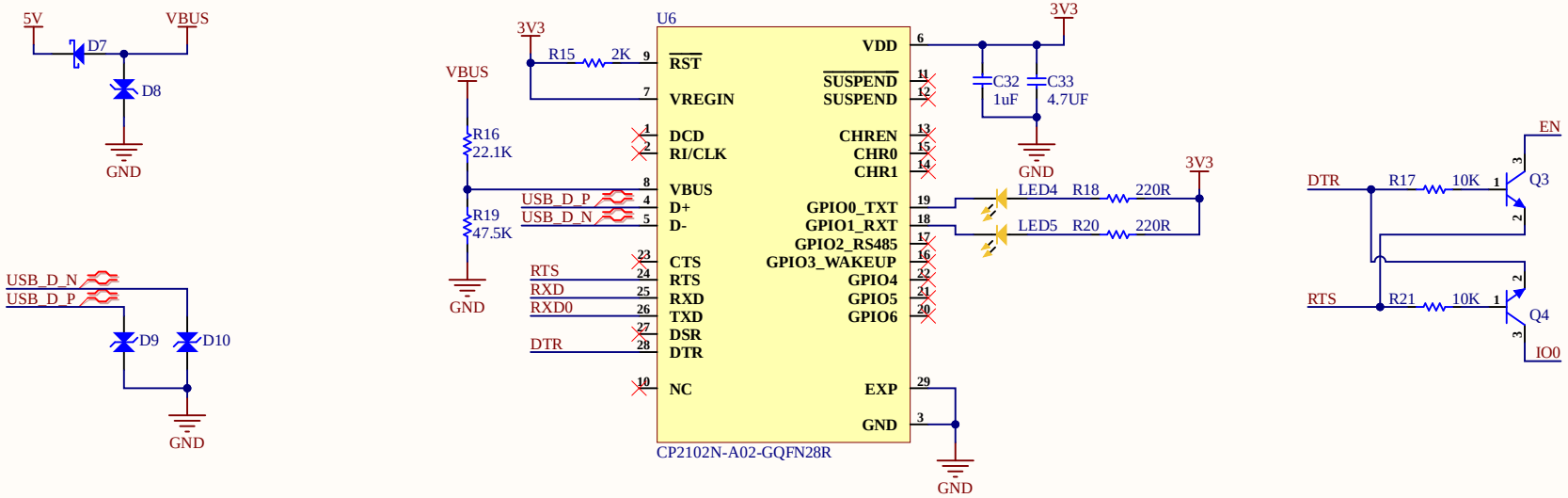


ISOLATED POWER SUPPLY



eSpark Consulting Group

Title: Power-Sheet-2.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 10

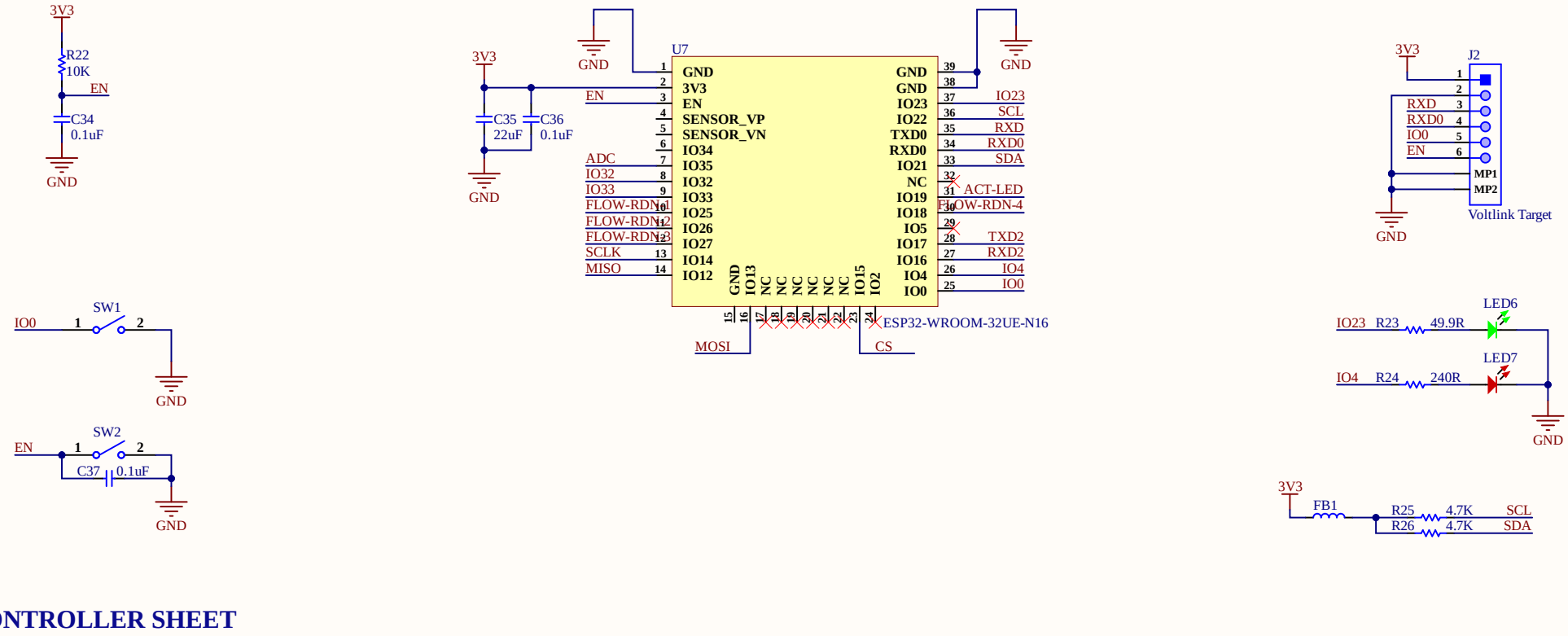


USB-TO-UART

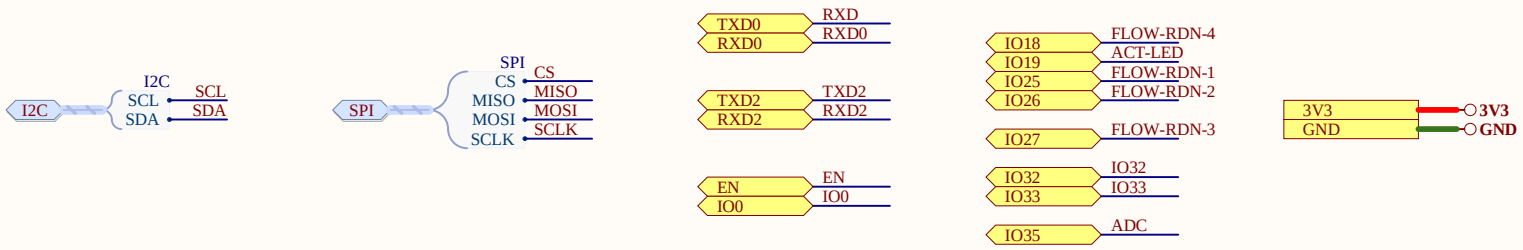
TXD
RXD
RXD0
RXD
EN
IO0
EN
IO0
USB
USB
USB_D_P
USB_D_N
USB_D_P
USB_D_N
VBUS
3V3
GND
VBUS
3V3
GND

eSpark Consulting Group

Title: USB-to-UART-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 10

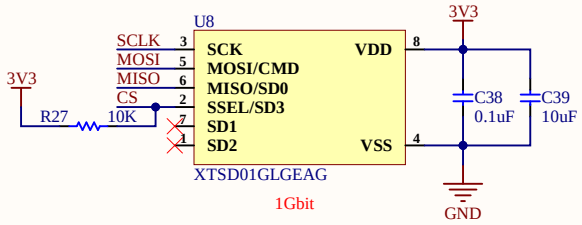


CONTROLLER SHEET



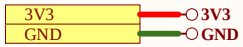
eSpark Consulting Group

Title: Controller-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 10



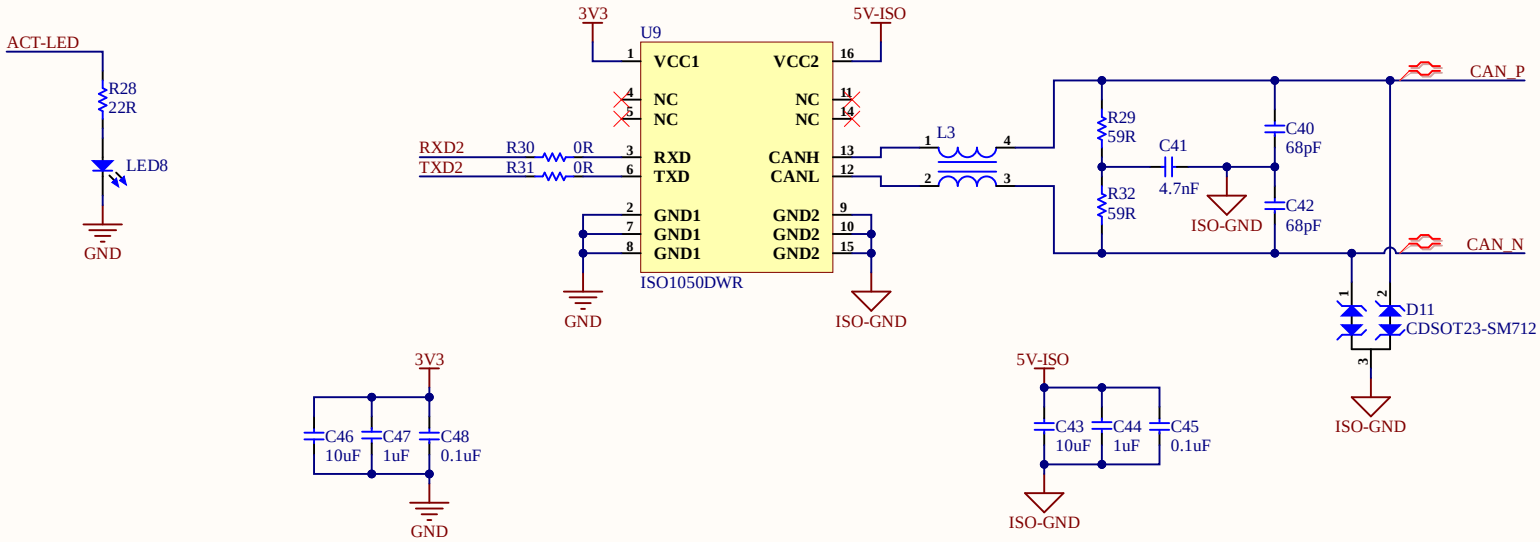
SD NAND





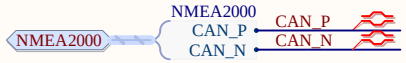
eSpark Consulting Group

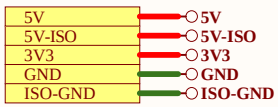
Title: Storage-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10



ISOLATED CAN TRANSCIEVER NMEA2000

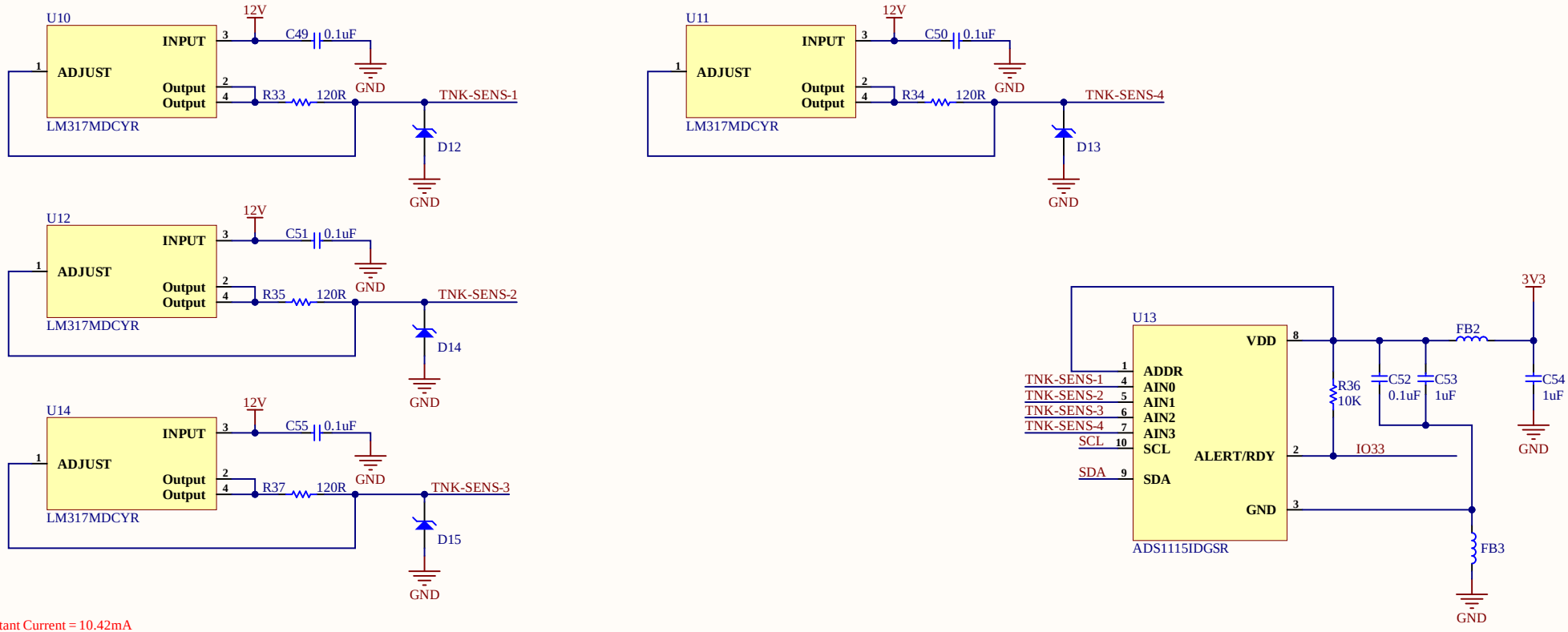






eSpark Consulting Group

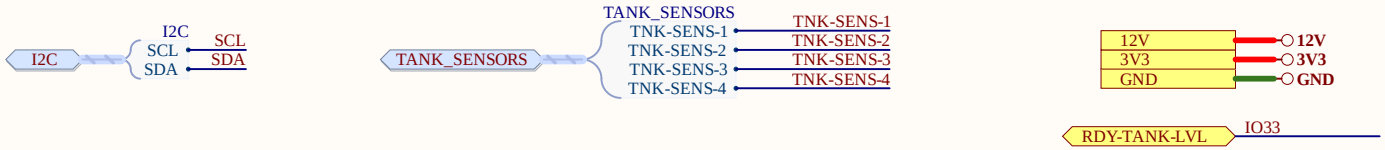
Title: CAN-Transceiver-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 10



Note:
Constant Current = 10.42mA

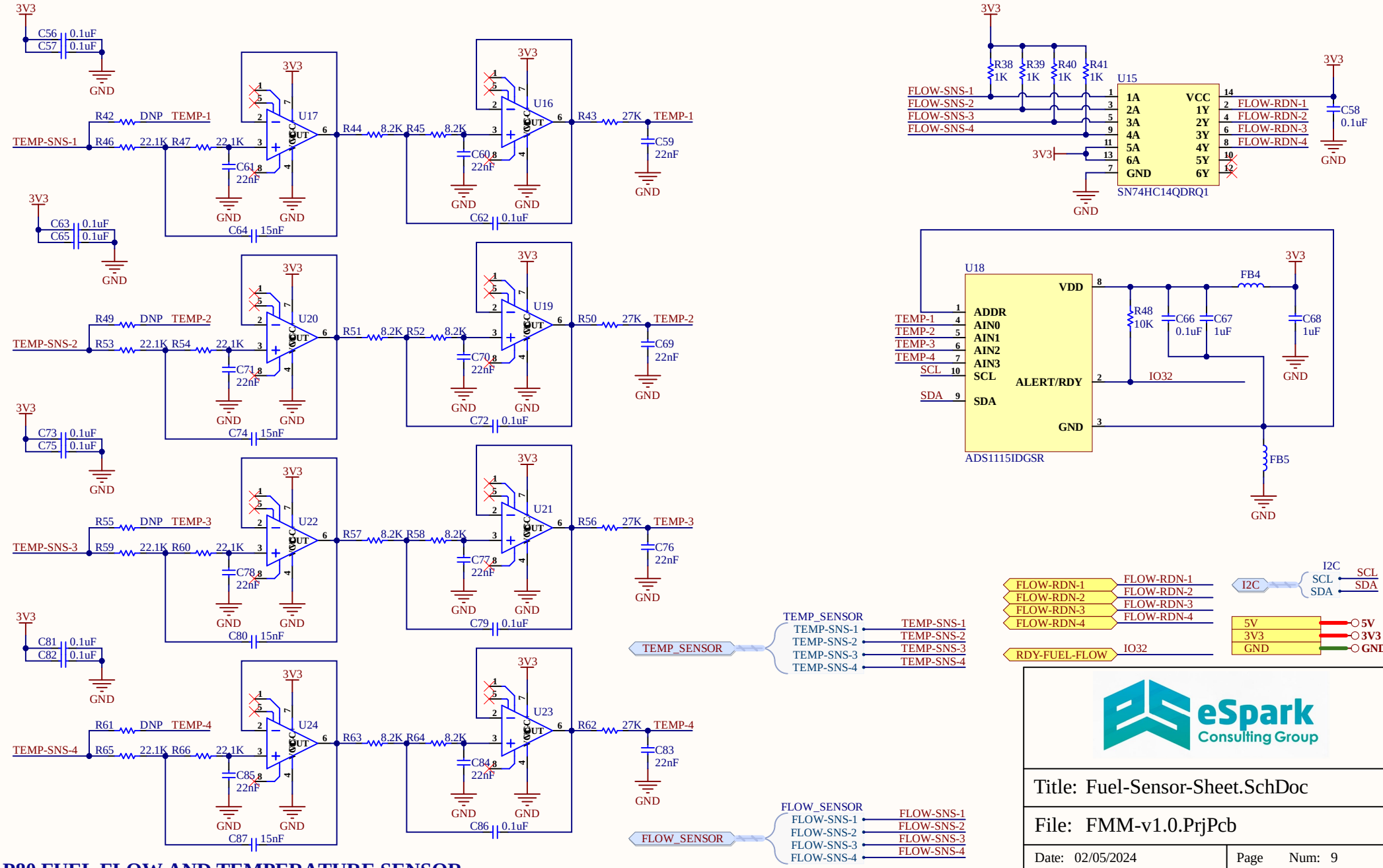
For 240 - 33Ω sensor:
Tank Empty Expected Value = 2.5V
Tank Full Expected Value = 0.34V

TANK-LEVEL SENSORS



eSpark Consulting Group

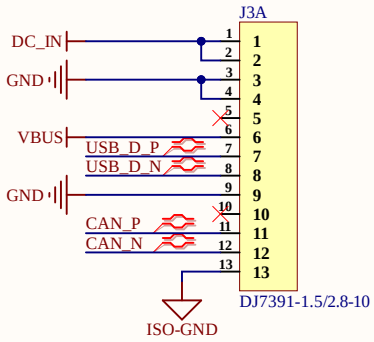
Title: Tank-Level-Sensor-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 10

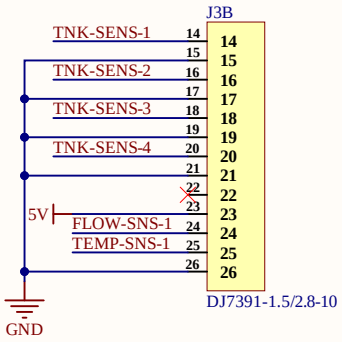


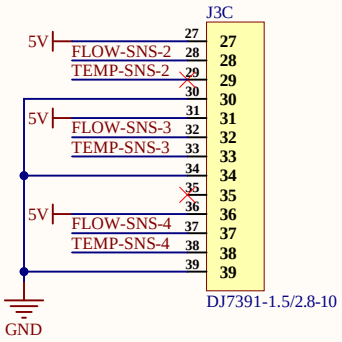
P80 FUEL FLOW AND TEMPERATURE SENSOR

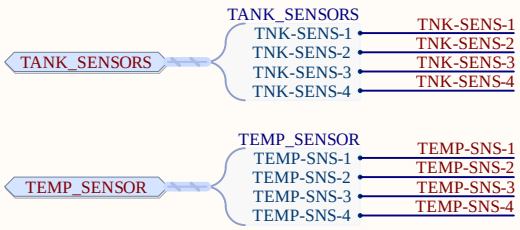
Title: Fuel-Sensor-Sheet.SchDoc	
File: FMM-v1.0.PrjPcb	
Date: 02/05/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 10

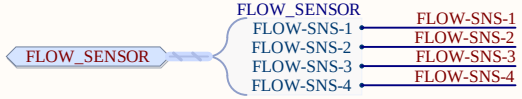
HARNESS CONNECTOR

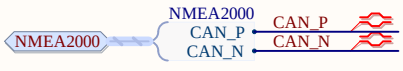




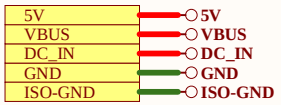












eSpark Consulting Group

Title: Connector-Sheet.SchDoc

File: FMM-v1.0.PrjPcb

Date: 03/05/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 10

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.200mm	4.1	
2	Power Layer	CF-004	0.035mm		
	Dielectric 1	FR-4	1.030mm	4.8	
3	Ground Layer	CF-004	0.035mm		
	Dielectric 3	PP-006	0.200mm	4.1	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.590mm		

